

GROWTH CONDITIONS TO ENSURE INPUT-TO-STATE STABILITY OF TIME-DELAY SYSTEMS WITH POINT-WISE DISSIPATION

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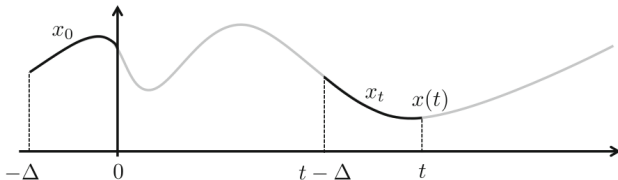
Time-delay system

Definition 1

A **time-delay system** (TDS) is a system modeled by:

$$\dot{x}(t) = f(x_t, u(t)). \quad (1)$$

$u \in L_{loc}^\infty(\mathbb{R}^+, \mathbb{R}^m)$, $\Delta > 0$, $x_t : [-\Delta, 0] \rightarrow \mathbb{R}^n$, $x_t(s) = x(t+s)$, $\forall s \in [-\Delta, 0]$. Assume f , Lipschitz on bounded sets and $f(0, 0) = 0$, in what follows.



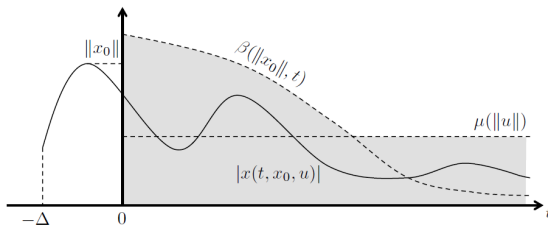
Input-to-state stability (ISS)

Definition 2

TDS (1) is said to be **input-to-state stable (ISS)** if there exist $\beta \in \mathcal{KL}$ and $\mu \in \mathcal{K}_\infty$ such that, for all $x_0 \in C([-\Delta, 0], \mathbb{R}^n)$ and all $u \in L_{loc}^\infty(\mathbb{R}^+, \mathbb{R}^m)$,

$$|x(t, x_0, u)| \leq \beta(\|x_0\|, t) + \mu(\|u_{[0,t]}\|), \quad \forall t \geq 0. \quad (2)$$

$$|x(t, x_0, u)| \leq \max\{\beta(\|x_0\|, t), \mu(\|u_{[0,t]}\|)\}, \quad \forall t \geq 0.$$



Definition 3

A functional $V : C([-\Delta, 0], \mathbb{R}^n) \rightarrow \mathbb{R}_{\geq 0}$ is said to be a **Lyapunov-Krasovskii functional candidate (LKF)** if it is **Lipschitz on bounded sets** and there exist $\underline{\alpha}, \overline{\alpha} \in \mathcal{K}_{\infty}$ such that, for all $t \geq 0$

$$\underline{\alpha}(|x(t)|) \leq V(x_t) \leq \overline{\alpha}(\|x_t\|). \quad (3)$$

$$|x(t)| := \left(\sum_{i=1}^n |x_i(t)|^2 \right)^{1/2} \quad \text{and} \quad \|x_t\| := \sup_{\tau \in [-\Delta, 0]} |x_t(\tau)|.$$

Definition 4

For TDS (1), an LKF V is said to be

- 1 an ISS LKF with **LKF-wise** dissipation if there exist $\alpha, \gamma \in \mathcal{K}_\infty$ such that

$$D^+V \leq -\alpha(V(x_t)) + \gamma(|u(t)|), \quad (4)$$

- 2 an ISS LKF with **point-wise** dissipation if there exist $\alpha, \gamma \in \mathcal{K}_\infty$ such that

$$D^+V \leq -\alpha(|x(t)|) + \gamma(|u(t)|). \quad (5)$$

D^+V is the Driver's derivative of the functional V along the solution of (1).

[Karafyllis et al., 2008]

Theorem 5

*TDS (1) is ISS if and only if it admits an ISS LKF with **LKF-wise dissipation**.*

Is the point-wise dissipation enough to ensure ISS of TDS?

[Chaillet et al., 2017]

Conjecture 1

Assume that the system (1) admits a LKF with a point-wise dissipation. Then it is ISS.

In response,

- 1 [Chaillet et al., 2017] provides a sufficient condition on the growth of the vector field of TDS.
- 2 [Chaillet et al., 2023] provides some growth conditions on the LKF to ensure exp-ISS.

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A way to get LKF-wise dissipation

Consider the LKF

$$V(x_t) = V_1(x(t)) + \int_{-\Delta}^0 V_2(x_t(s))ds, \quad (6)$$

which dissipates point-wisely as follow

$$D^+V \leq -\alpha(|x(t)|) + \gamma(|u(t)|).$$

By adding ke^{cs} in the kernel of integral part of (6), we get the LKF

$$W(x_t) = V_1(x(t)) + \int_{-\Delta}^0 ke^{cs} V_2(x_t(s))ds.$$

Proposition 1

If $\alpha(|x(t)|) \geq pV_2(x(t))$, ($p > 0$) then W is a LKF with LKF-wise dissipation and the TDS (1) is ISS.

Corollary: $(V_1, V_2, \alpha \text{ quadratics}) = [\text{Orłowski et al., 2022, Lemma 1}]$.

Does this trick work systematically?

Consider the following **1D** TDS

$$\dot{x}(t) = -x(t) - \frac{x(t)}{1+x(t)^2} + \frac{x(t-1)^4}{1+|x(t)|^3} + \frac{u(t)}{1+x(t)^2}, \quad (7)$$

and the LKFs:

$$V(x_t) := \frac{1}{4}x(t)^4 + \int_{-1}^0 x_t(s)^4 ds,$$

$$W(x_t) := \frac{1}{4}x(t)^4 + \int_{-\Delta}^0 \textcolor{red}{ke^{cs}} x_t(s)^4 ds.$$

Proposition 2

- 1 System (7) **is ISS**.
- 2 V is an ISS LKF with **point-wise dissipation** for (7)
- 3 Given any $k, c > 0$, W **is not a LKF with LKF-wise dissipation**.

The ISS of (7) is shown using Razumikhin method ([Teel, 1998]).

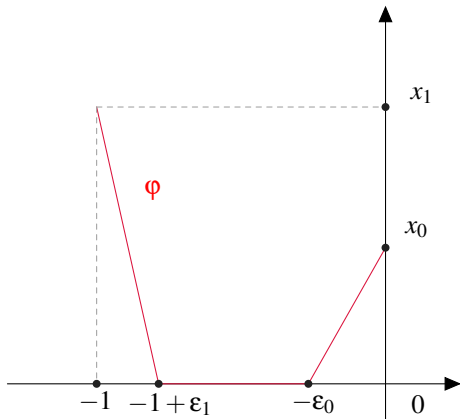
Sketch of the proof

Here $\Delta = 1$, we set $\mathcal{X} := C^0([-1, 0], \mathbb{R})$ and we denote by f the dynamic of (7).

W dissipates LKF-wise $\Rightarrow D_0^+ W := D^+ W(\phi, f(\phi, 0)) \leq -\alpha(V(\phi)) \leq 0, \quad \forall \phi \in \mathcal{X}.$

$$D_0^+ W = x_0^4 \left(k - 1 - \frac{1}{1 + x_0^2} \right) + x_1^4 \left(\frac{x_0^3}{1 + |x_0|^3} - k e^{-c} \right) - kc \int_{-1}^0 e^{cs} \phi(s)^4 ds.$$

$$D^+ W(\phi, f(\phi, 0)) > 0.$$



Theorem 6

Assume that there exists a LKF V which dissipates *point-wisely* as follow,

$$D^+V \leq -\alpha(Q(x(t))) + \gamma(|u(t)|). \quad (8)$$

Assume that

$$\dot{Q}(x(t)) \leq \sigma(\|Q\|) + \gamma(|u(t)|). \quad (9)$$

Then, if

$$\liminf_{r \rightarrow +\infty} \frac{\alpha(r)}{\sigma(re^{2\Delta})} > 0, \quad (10)$$

the system (1) is ISS.

Corollary 7

Assume that there exists a LKF V which dissipates *point-wisely* as follow,

$$D^+V \leq -Q(x(t)) + \gamma(|u(t)|). \quad (11)$$

Assume that

$$\dot{Q}(x(t)) \leq \sigma(\|Q\|) + \gamma(|u(t)|). \quad (12)$$

Then, if

$$\liminf_{r \rightarrow +\infty} \frac{r}{\sigma(r)} > 0, \quad (13)$$

the system (1) is ISS.

Conclusion

- 1 The principle of adding exponential term in relaxed LKFs to make them strict, does not systematically work even for $1D$ TDS.
- 2 Growth condition is proposed to conclude ISS with relaxed LKF.
- 3 The proposed condition turns out to extend the existing ones.
- 4 The conjecture is still an open question.



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